SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario Based Assignment

Code of Conduct:

I V G ABBIRAMY -192524097 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V G ABBIRAMY

Annexure A

Scenario Based Assignment

1. **Scenario** : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

i. **Greedy Best-First Search (GBFS)**

Greedy Best-First Search selects the next node based only on the heuristic value **h(n)** (estimated distance to the goal).

Strengths:

- Very fast in finding a route.
- Suitable for emergency situations where quick decisions are needed.
- Uses less computation than A*.

Weaknesses:

- Ignores the actual travel cost.
- May choose blocked or risky paths.
- Does not always find the optimal route.

ii. A* Search

A* uses:

$$f(n)=g(n)+h(n)$$

Where:

- **g(n)** = actual travel cost
- **h(n)** = estimated distance to goal

A* balances both actual cost and heuristic estimate, producing safer and more optimal routes.

iii. Impact of Heuristic Accuracy

- **Accurate heuristic:** Both GBFS and A* perform efficiently.
- **Poor heuristic:** GBFS may take wrong paths, while A* still considers actual cost and usually finds a better solution.
- **Admissible heuristic:** Guarantees optimality in A*.

iv. Effect of Dynamic Changes

Blocked roads or weather changes require the search algorithm to replan.

- GBFS may repeatedly select blocked paths.
- A* updates path costs and searches for an alternative optimal path.

v. Recommended Algorithm

*A Search**

Justification:

- Produces optimal paths.
- Considers travel cost and heuristic.
- Handles dynamic environments better.
- Provides safer navigation for emergency medicine delivery.

2. **Scenario**: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks :

Problem Formulation

Objective:

- Minimize waiting time
- Reduce fuel consumption
- Reduce traffic congestion

Variables:

- Traffic signal timings

Constraints:

- Dynamic traffic flow
- Large search space

Traditional search is inefficient because exploring every possible signal combination requires enormous computation.

i. Hill Climbing

Hill Climbing starts with an initial signal timing and continuously moves to a better neighboring solution.

Advantages

- Simple
- Fast
- Low memory requirement

Limitations

- Can stop at local optimum.
- Cannot explore worse states to reach better solutions.

ii. Local Maxima, Plateaus and Ridges

Local Maximum

- Traffic improves slightly but not globally.

Plateau

- Several signal timings produce similar results, causing no improvement.

Ridge

- Best solution requires multiple coordinated timing changes, which Hill Climbing struggles to find.

iii. Simulated Annealing / Genetic Algorithm

Simulated Annealing

- Occasionally accepts worse solutions.
- Escapes local maxima.

Genetic Algorithm

- Maintains multiple candidate solutions.
- Uses selection, crossover, and mutation.
- Efficient for very large optimization problems.

iv. Recommended Approach

Genetic Algorithm

Justification

- Highly scalable.
- Adapts to changing traffic conditions.
- Produces high-quality solutions for large search spaces.

3. **Scenario:** An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

i. Why Online Search?

The rover has no complete map and discovers obstacles while moving.

Offline search requires complete knowledge before planning, making it unsuitable.

ii. Challenges

- Partial observability
- Unknown obstacles
- Limited sensing
- Dynamic environment
- Communication delay

iii. Internal Model

The rover:

1. Starts with limited information.
2. Observes nearby cells.
3. Updates its internal map.
4. Avoids hazards.
5. Replans whenever new information is available.

iv. Comparison

Online Search	Uninformed Search	Informed Search
Learns while moving	No heuristic	Uses heuristic
Works in unknown environments	Requires known environment	Mostlyrequires map
Adaptive	Less adaptive	Faster but assumes prior knowledge

v. Recommended Approach

Online Search Agent

Justification

- Suitable for unknown terrain.
- Continuously updates knowledge.
- Adapts to changing conditions.
- Ensures safer navigation.

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

CSP Formulation

Variables

- Subjects
- Exam halls
- Time slots

Domains

- Available halls
- Available dates
- Available time slots

Constraints

- No overlapping exams.
- Limited halls.
- Faculty availability.
- Subject precedence.
- Special accommodations.

i. Variables, Domains and Constraints

Variables: Exams

Domains: Possible time slots and halls

Constraints:

- One hall per exam.
- No student conflict.
- Faculty availability.
- Special student requirements.
- Subject ordering.

ii. Backtracking Search

1. Assign a time slot.
2. Check constraints.
3. If constraints fail, backtrack.
4. Try another assignment.
5. Continue until all exams are scheduled.

iii. Constraint Propagation (Forward Checking)

Forward checking removes invalid future assignments after every allocation.

Benefits:

- Reduces search space.
- Detects conflicts early.
- Improves efficiency.

iv. Best Strategy

Backtracking + Forward Checking

Justification

- Finds valid schedules efficiently.
- Reduces unnecessary computation.
- Scales well for many courses and students.

5. Scenario: Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

i. Minimax Algorithm

Minimax assumes:

- AI maximizes score.
- Opponent minimizes score.

The algorithm evaluates future game states and selects the move with the highest utility.

ii. Computational Challenges

- Very large game tree.
- High branching factor.
- Large memory usage.
- Slow decision making.

iii. Alpha-Beta Pruning

Alpha-Beta pruning removes branches that cannot affect the final decision.

Benefits:

- Evaluates fewer nodes.
- Produces the same optimal result as Minimax.
- Reduces computation time.
- Suitable for real-time games.

iv. Evaluation Function

Example:

Evaluation Score = Material Advantage + Position Score + Mobility − Risk

Factors:

- Number of pieces
- Board position
- Possible legal moves
- Threat level
- King safety

v. Recommended Strategy

Minimax with Alpha-Beta Pruning

Justification

- Produces near-optimal decisions.
- Significantly reduces computation.
- Meets real-time constraints.
- Widely used in modern game AI.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand